

Class 09: Candy Mini Project

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Table of contents

Importing the Candy Data	1
What is in the dataset?	2
What is your favorite candy?	2
Exploratory Analysis	5
Overall Candy Rankings	7
Taking a look at pricepercent	14
Exploring the correlation structure	15
Principal Component Analysis	16
Summary	21

Importing the Candy Data

```
#Load the dataset
candy <- read.csv("candy-data.txt", row.names = 1)

head(candy)
```

	chocolate	fruity	caramel	peanut	almondy	nougat	crisped	ricewafer
100 Grand	1	0	1		0	0		1
3 Musketeers	1	0	0		0	1		0
One dime	0	0	0		0	0		0
One quarter	0	0	0		0	0		0
Air Heads	0	1	0		0	0		0
Almond Joy	1	0	0		1	0		0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294

One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

What is in the dataset?

```
#Nrow to see rows, sum to sum up the T/F
nrow(candy)
```

```
[1] 85
```

```
sum(candy$fruity)
```

```
[1] 38
```

Q1. How many different candy types are in this dataset?

There are 85 different candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

There are 38 different fruity candy types in the dataset.

What is your favorite candy?

```
#Load the dplyr library
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

#We filter it to a singular row with the filter, and then select a specific area with select

```
candy|>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

```
candy|>
  filter(row.names(candy)=="Snickers") |>
  select(winpercent)
```

```
      winpercent
Snickers  76.67378
```

```
candy|>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat   76.7686
```

```
candy|>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars  49.6535
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy is Snickers and it's winpercent value is 76.67378.

Q4. What is the winpercent value for "Kit Kat"?

The winpercent value for Kit Kat is 76.7686.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The winpercent value for Tootsie Roll Snack Bars is 49.6535.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Winpercent is the only one that isn't on a 0-1 scale.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

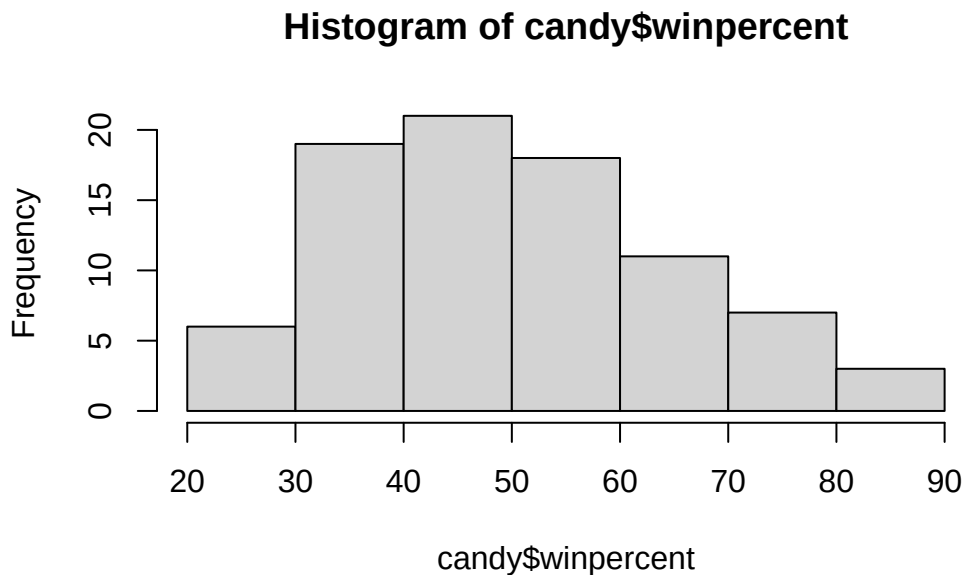
One means that it is true that the candy does contain chocolate, a zero means it does not.

Exploratory Analysis

```
#Load ggplot  
library(ggplot2)
```

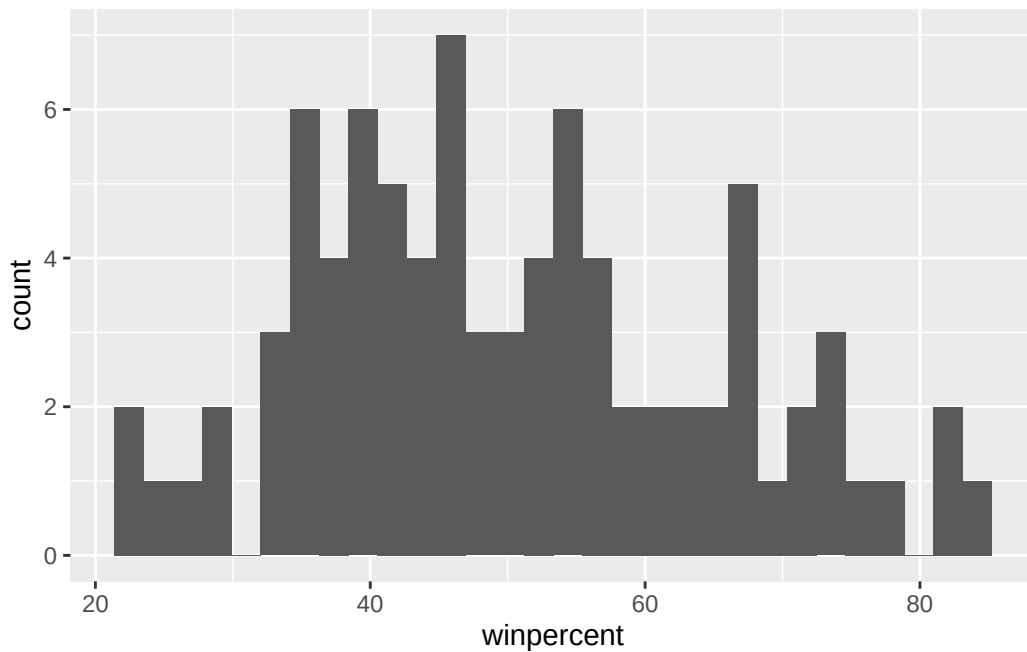
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
#Default R histogram  
hist(candy$winpercent)
```



```
#ggplot histogram  
ggplot(candy)+  
  aes(winpercent)+  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution of winpercent values are not symmetrical.

Q10. Is the center of the distribution above or below 50%?

It is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average chocolate is higher ranked than fruit candy.

```
fruitywin <- mean(candy$winpercent[as.logical(candy$fruity)])
chocowin <- mean(candy$winpercent[as.logical(candy$chocolate)])

fruitywin > chocowin
```

[1] FALSE

Q12. Is this difference statistically significant?

Yes, this difference is statistically significant.

```
t.test(candy$winpercent[as.logical(candy$chocolate)], candy$winpercent[as.logical(candy$fruity)], var.equal = FALSE)
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Overall Candy Rankings

```
head(candy[order(candy$winpercent, decreasing = TRUE),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1

	crispedricewafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0.720
Reese's Miniatures	0	0	0	0.034
Twix	1	0	1	0.546
Kit Kat	1	0	1	0.313
Snickers	0	0	1	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

```
head(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q13. What are the five least liked candy types in this set?

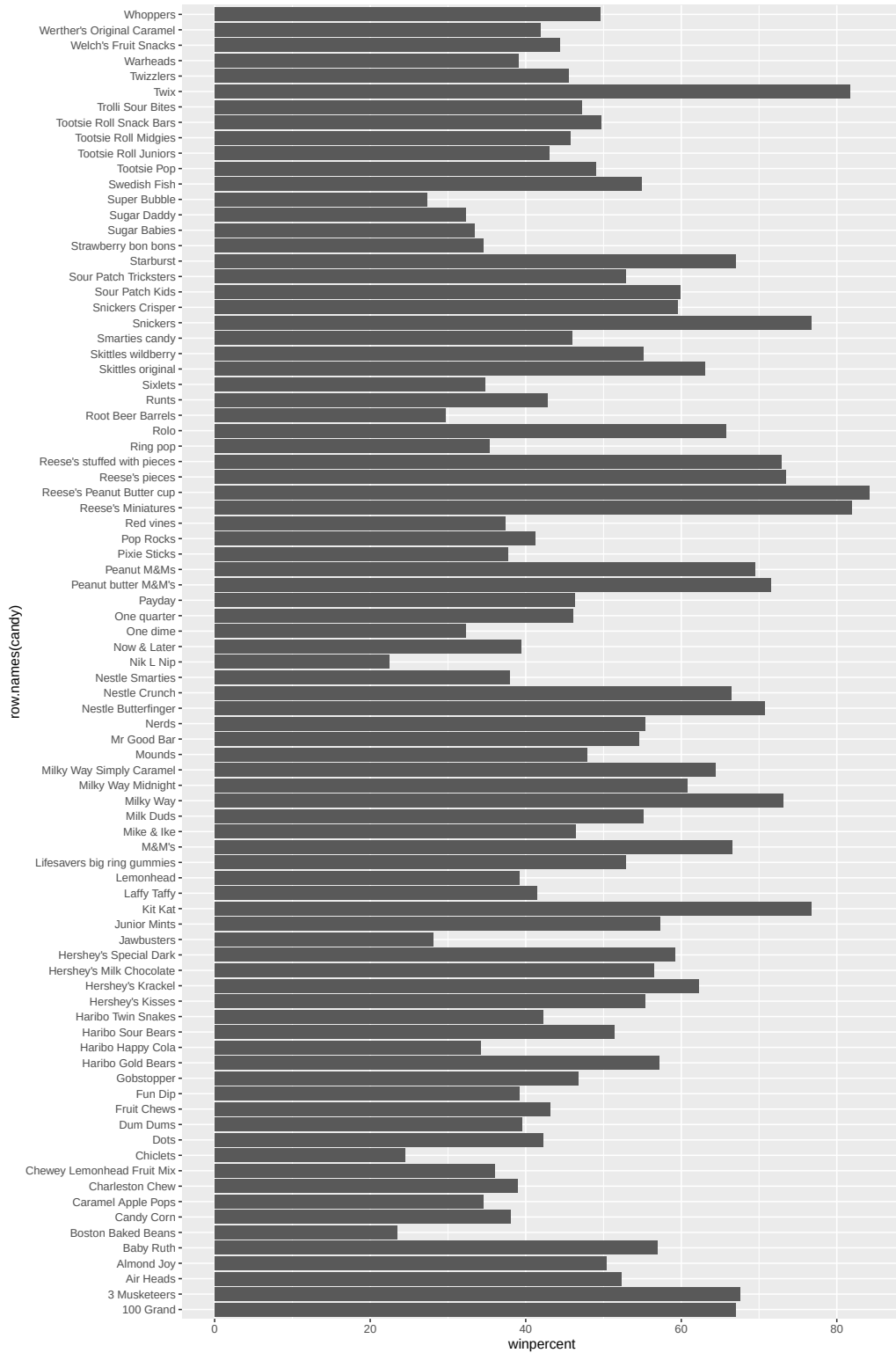
The 5 least liked candy types in this data set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

The top 5 all time candy types are Reese's Peanut Butter Cups, Reese's Miniatures, Twix, Kit Kat and Snickers.

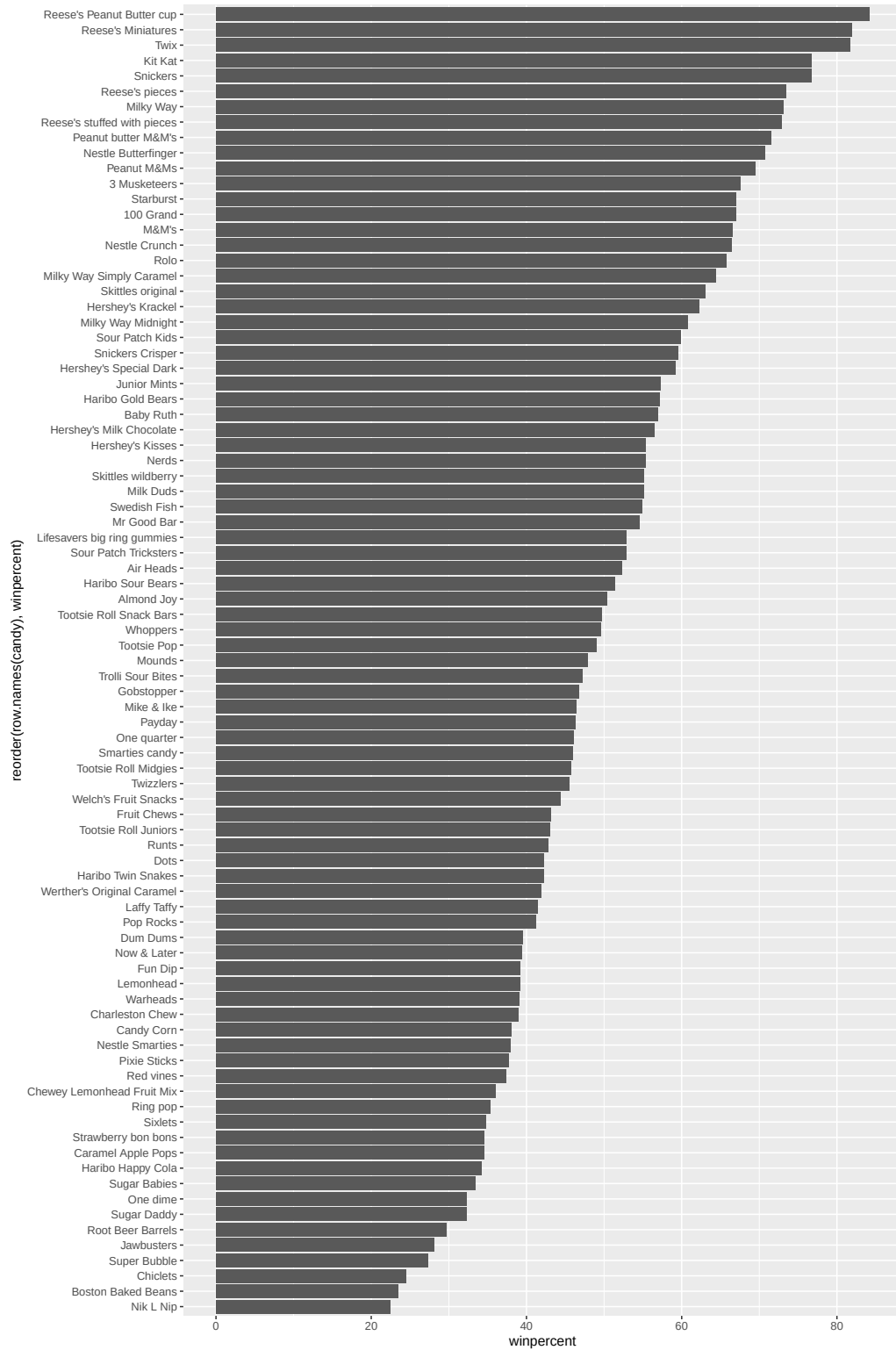
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy)+
  aes(winpercent, row.names(candy))+
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy)+  
  aes(winpercent, reorder(row.names(candy), winpercent))+  
  geom_col()
```



```
my_cols=rep("black", nrow(candy))  
my_cols[as.logical(candy$chocolate)] = "chocolate"  
my_cols[as.logical(candy$bar)] = "brown"  
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy),winpercent)) +  
  geom_col(fill=my_cols) +  
  ylab("")
```


Q17. What is the worst ranked chocolate candy?

Sixlets is the worst ranked chocolate candy.

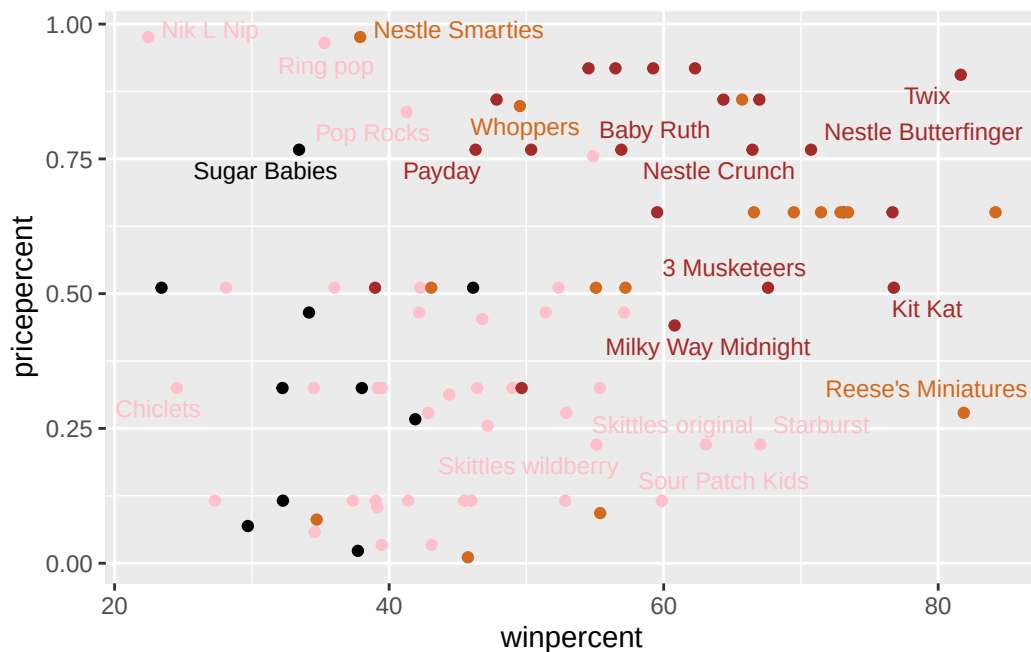
Q18. What is the best ranked fruity candy?

Starburst is the best ranked fruity candy.

Taking a look at pricepercent

```
library(ggrepel)
```

```
ggplot(candy) +  
  aes(winpercent, pricepercent, label=rownames(candy)) +  
  geom_point(col=my_cols) +  
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures seem to be the most bang for your buck looking at the graph above.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, Hershey's Milk Chocolate. The least popular of these is Nik L Nip.

```
candy |>
  arrange(-pricepercent) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

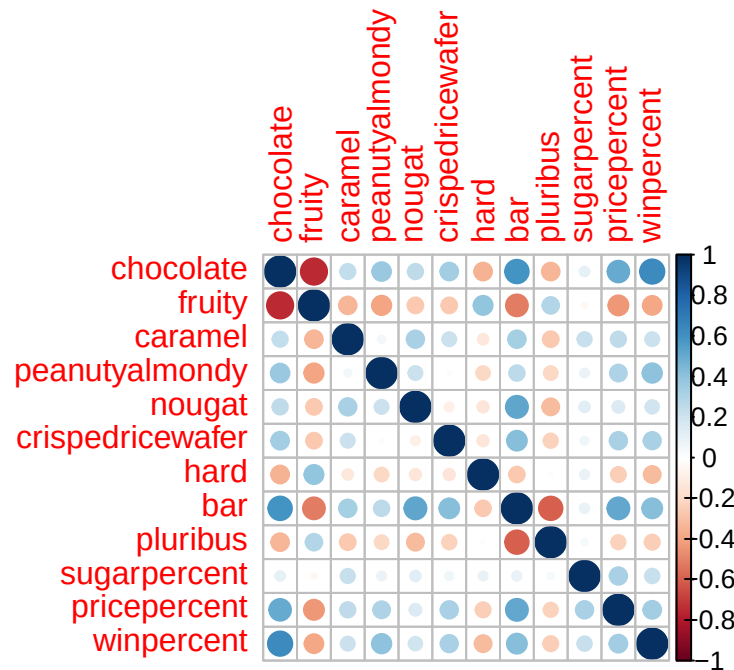
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

From the corplot, it seems the two variables that are the most anti-correlated are chocolate and fruity.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Principal Component Analysis

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

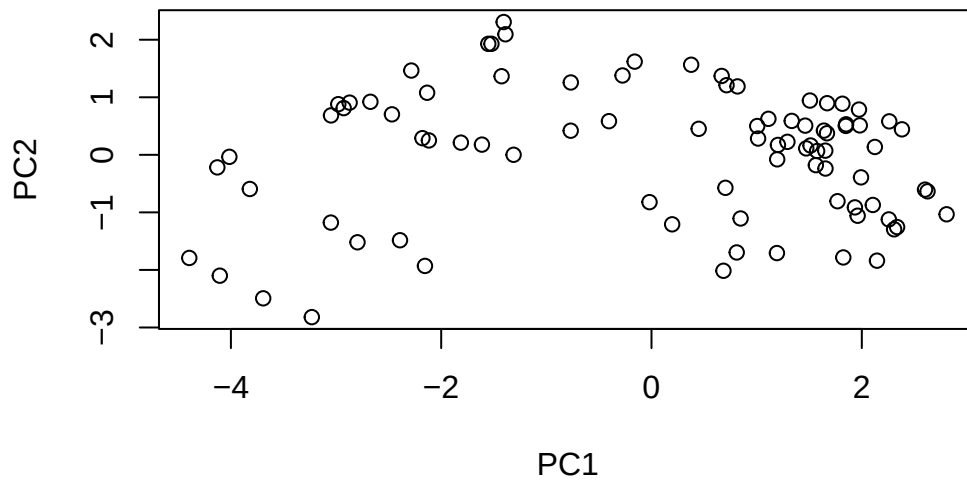
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

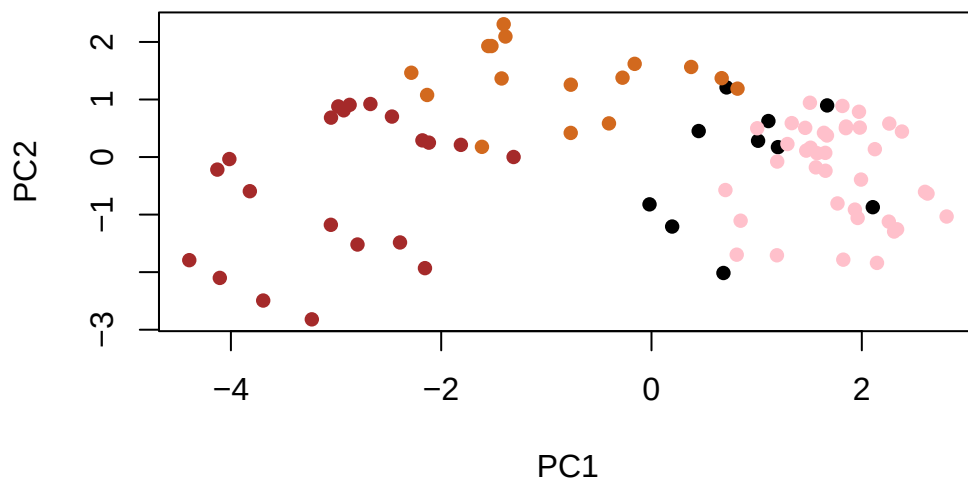
	PC8	PC9	PC10	PC11	PC12
--	-----	-----	------	------	------

Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



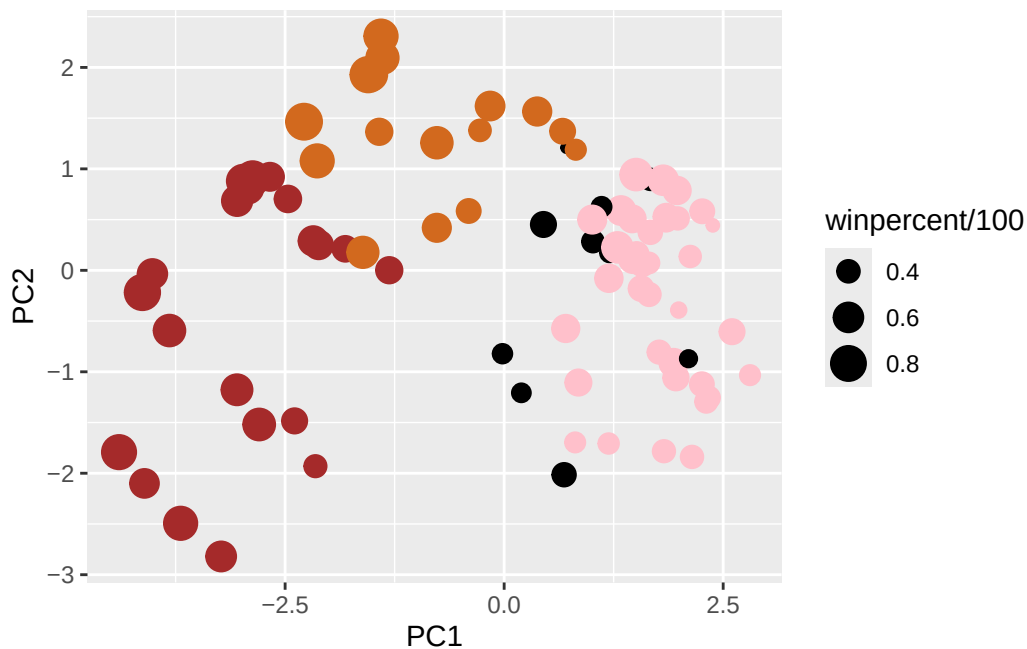
```
my_data <- cbind(candy, pca$x[,1:3])
head(my_data)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0
Air Heads	0	1	0		0	0			0
Almond Joy	1	0	0		1	0			0
	hard	bar	pluribus	sugar	percent	price	percent	win	percent
100 Grand	0	1	0	0.732	0.860	66.97173	-3.8198617		
3 Musketeers	0	1	0	0.604	0.511	67.60294	-2.7960236		
One dime	0	0	0	0.011	0.116	32.26109	1.2025836		
One quarter	0	0	0	0.011	0.511	46.11650	0.4486538		
Air Heads	0	0	0	0.906	0.511	52.34146	0.7028992		
Almond Joy	0	1	0	0.465	0.767	50.34755	-2.4683383		
	PC2	PC3							
100 Grand	-0.5935788	-2.1863087							
3 Musketeers	-1.5196062	1.4121986							
One dime	0.1718121	2.0607712							
One quarter	0.4519736	1.4764928							

Air Heads	-0.5731343	-0.9293893
Almond Joy	0.7035501	0.8581089

```
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

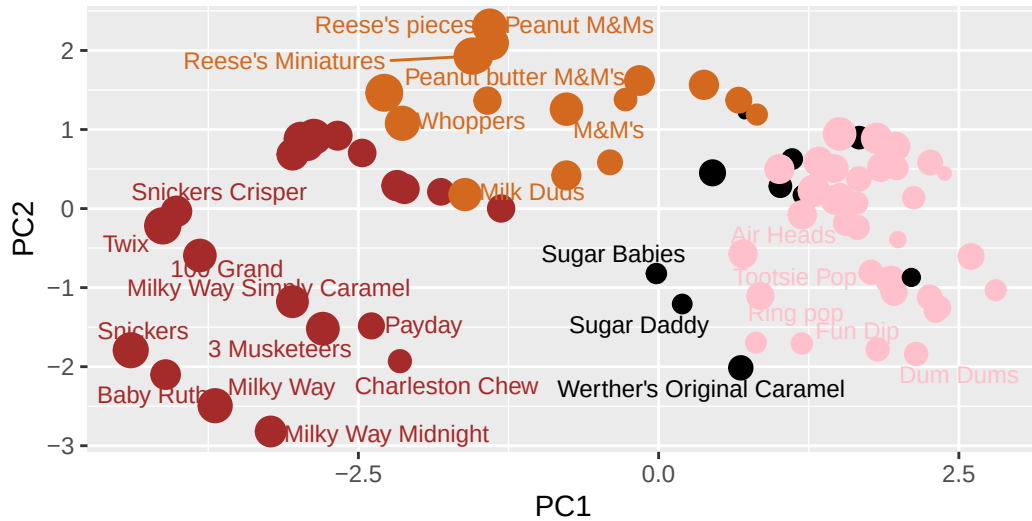
p



```
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

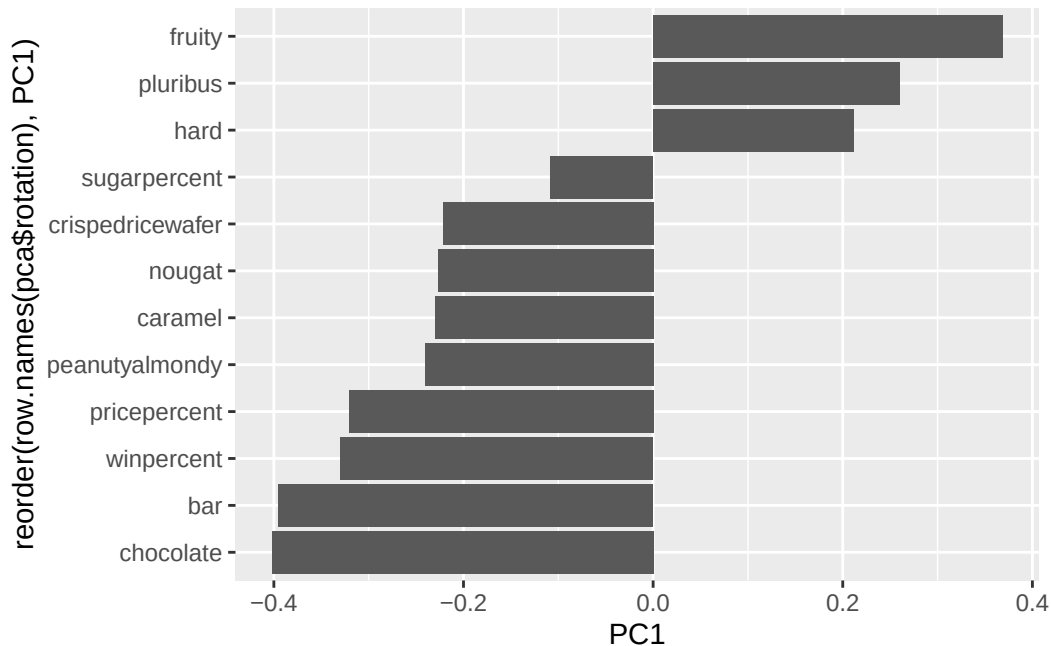
filter

The following object is masked from 'package:graphics':

layout

```
#ggplotly(p)
```

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The three variables picked up strongly in the positive direction are fruity, pluribus, and hard. This does make sense as the three were slightly unique in the way that they had a negative correlation with most everything else, but positive with each other.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A winning candy is a chocolate bar with peanuts with an okay price point. Exploratory analysis helps with identifying that chocolate was statistically significant. Correlation analysis shows which traits cluster together. Then PCA gives us the structure and confirms what is driving the variance. It helps explain the previous two.